DS	**Session**	Graph	**Question Information**	Level 1 Challenge 87

Question description

There are n cities and initially no roads between them. However, every day a new road will be constructed, and there will be a total of m roads.

A component is a group of cities where there is a route between any two cities using the roads. After each day, your task is to find the number of components and the size of the largest component.

Input

The first input line has two integers n and m: the number of cities and roads. The cities are numbered 1,2,…,n.

Then, there are m lines describing the new roads. Each line has two integers a and b: a new road is constructed between cities a and b.

You may assume that every road will be constructed between two different cities.

Output

Print m lines: the required information after each day.

Constraints

- $1 \le n \le 10^5$
- $1 \le m \le 2 \cdot 10^5$
- $1 \le a,b \le n$

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 6
1 2
2 3
3 1
3 4
4 5
5 4
```

EXPECTED OUTPUT

```
4 2
3 3
3 3
2 4
1 5
1 5
```

Test Case 2

INPUT (STDIN)

```
5 5
1 3
2 4
3 1
3 2
5 4
```

EXPECTED OUTPUT

```
4 2
3 2
3 2
2 4
1 5
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
int join(int i,int j)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
2
```

Test Case 2

TOKEN COUNT

```
300
```

Test Case 3

NLOC

```
40
```

Code

road_construction.c

```c
#include <stdio.h>
#include <stdlib.h>

#define MAXN 100005

int parent[MAXN];
int sz[MAXN];
int num_components;
int max_component_size = 1;

int find_rep(int v) {
    if (v == parent[v]) return v;
    return parent[v] = find_rep(parent[v]);
}

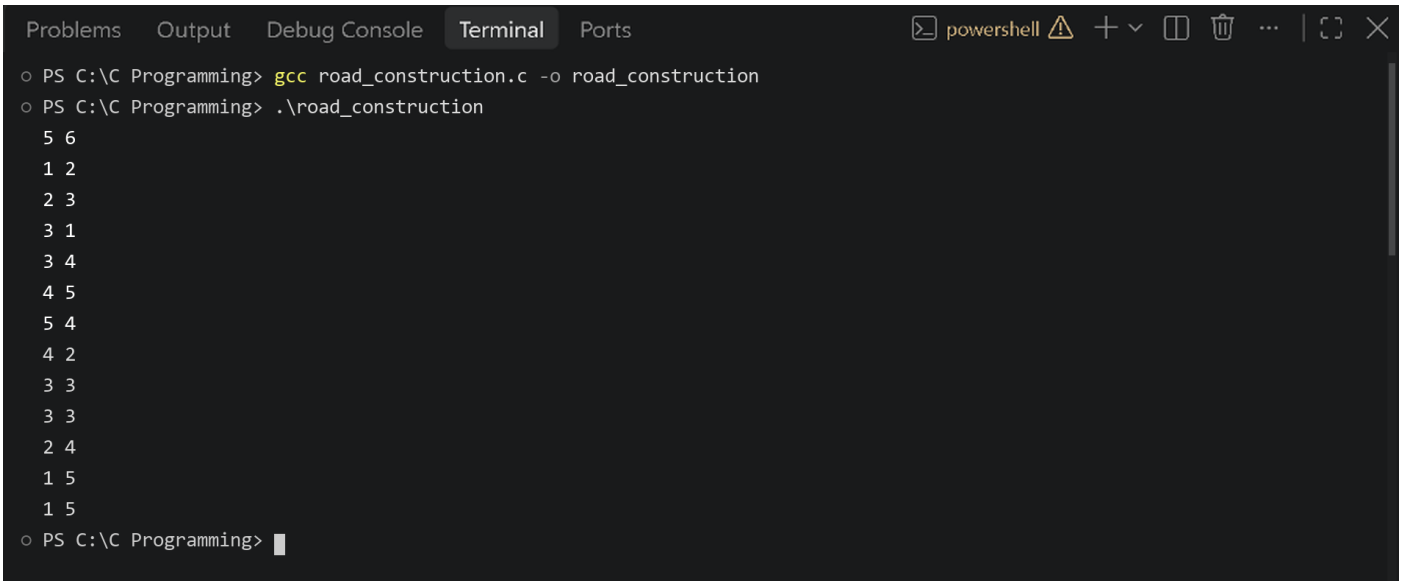
int join(int i, int j) {
    int root_i = find_rep(i);
    int root_j = find_rep(j);
    if (root_i != root_j) {
        if (sz[root_i] < sz[root_j]) {
            int t = root_i; root_i = root_j; root_j = t;
        }
        parent[root_j] = root_i;
        sz[root_i] += sz[root_j];
        if (sz[root_i] > max_component_size) {
            max_component_size = sz[root_i];
        }
        num_components--;
    }
    return num_components;
}

int main() {
    int n, m;
    if (scanf("%d %d", &n, &m) != 2) return 0;

    num_components = n;
    for (int i = 1; i <= n; i++) {
        parent[i] = i;
        sz[i] = 1;
    }

    for (int e = 0; e < m; e++) {
        int u, v;
        scanf("%d %d", &u, &v);
        join(u, v);
        printf("%d %d\n", num_components, max_component_size);
    }
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc road_construction.c -o road_construction
PS C:\C Programming> .\road_construction
5 6
1 2
2 3
3 1
3 4
4 5
5 4
4 2
3 3
3 3
2 4
1 5
1 5
PS C:\C Programming>